



Evaluation of Shoulder Pain

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Why Shoulder Pain Matters

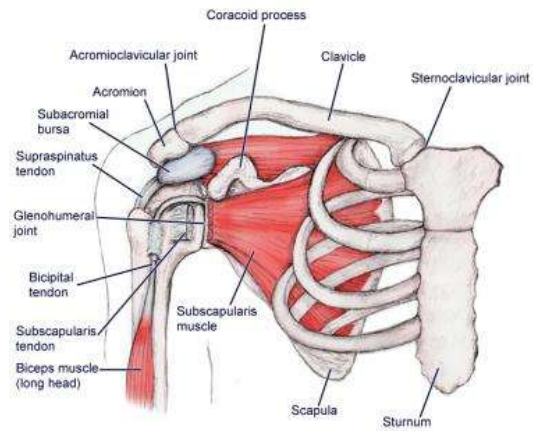
- Very common complaint, especially in active adults

- Usually mechanical but can be systemic or referred

- Key goal: localize the pain and determine the structure involved

Again, a common presentation that affects many folks, not just athletes!

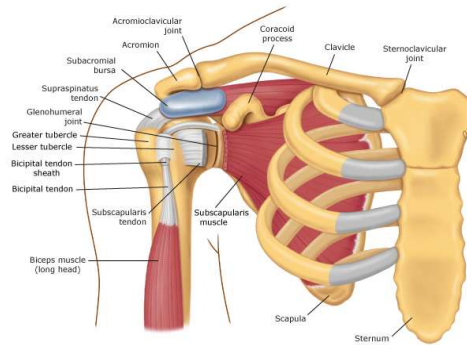
Shoulder:
know your
anatomy!



3 broad categories of differential

- Soft Tissue
- Orthopedic Structures
- Neurologic Structures

Anterior view of shoulder anatomy



Graphic 72709 Version 3.0

Common Causes

Soft Tissue – tendinopathy, bursitis, partial or full tear

Structural/Joint – Arthritis, adhesive capsulitis, instability

Traumatic – Dislocation, fracture, labral tear

Referred – Cervical spine, cardiac, diaphragmatic sources

Focused history

- Remember your “OLD CARTS”
 - Onset
 - Location/Radiation
 - Duration
 - Character
 - Aggravating factors
 - Relieving factors
 - Timing
 - Severity
- Trauma
- Inciting event
- Weakness
- Similar problem in the past
- Specific limitation of movements
- Pain with certain movements
- Therapies tried
- Affect on ADLs

Focused History: Shoulder Pain

◆ Onset & Mechanism

- Acute trauma (fall, dislocation) vs insidious (overuse, degeneration)
- Overhead or repetitive work?

◆ Location of Pain

- Lateral deltoid region → rotator cuff
- Anterior shoulder/groove → biceps tendon
- Superior shoulder → AC joint
- Deep joint pain → labral pathology

◆ Quality & Timing

- Dull ache, worse at night → rotator cuff
- Sharp pain with overhead activity → impingement
- Instability, “slipping out” → dislocation/laxity

◆ Associated Symptoms

- Weakness (tear vs pain inhibition)
- Clicking/popping (labral)
- Stiffness (adhesive capsulitis)

◆ Red Flags

- History of malignancy

- Fever, weight loss, night sweats
- Acute trauma with obvious deformity

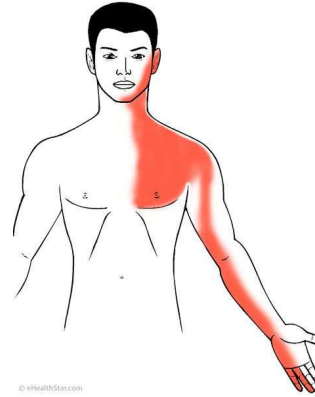
Related History

- PMH
 - Osteoarthritis, Rheumatoid arthritis, other autoimmune disease, gout, systemic illness
- PSH
 - Orthopedic surgeries, any other recent surgery (think referred pain)
- FH
 - Arthritis, gout, autoimmune disease
- SH
 - Work, hobbies, history of STI, current or past IV drug use

Referred pain (and red flags)

- Cervical spondylosis
 - Myocardial ischemia/infarction
 - Diaphragmatic irritation
 - Gallbladder disease
-
- If you suspect these, make sure you pay attention to vitals, constitutional symptoms, cardiac and GI history

Heart-Related Pain



VERY IMPORTANT !!! Always consider red flags that could present like an MSK issue but it is referred from another system. Make sure your history matches your physical and don't ignore concerning symptoms. Ex) the patient who presents with L shoulder pain but also seems diaphoretic, has some mild chest pressure and looks uncomfortable. They have cardiac risk factors, avoid tunnel vision and consider that a cardiac etiology like MI can play a role.

Osteopathic Exam

- Test for somatic dysfunction in upper thoracic (T2 – T5) region for:
 - Tissue texture change
 - Asymmetry
 - Range of Motion
 - Tenderness

- This upper thoracic region is where upper extremity SD may be found on structural exam.



Shoulder PE Overview

- Inspection
- Palpation
- Range of motion
- Strength
- +/- Neurovascular
- Special tests
- Special Tests
 - Intrinsic/Rotator Cuff
 - Impingement
 - Extrinsic
- OPP considerations
 - Clavicle, Scapulothoracic
 - Articulation slide/glide



Inspection

- Inspect both anterior and posterior shoulders
- Compare shoulder heights for symmetry
- Compare contour of shoulders and shoulder girdle
- Assess carrying angle
- Note any swelling, deformity, muscle atrophy, or tremors

Palpation

- Palpate and assess for pain
- Start at sternoclavicular joint
- Palpate along clavicle
- Feel acromioclavicular joint
- Subacromial bursa region
- Extend arm to palpate the rotator cuff muscles
- Long head of biceps tendon



Traction

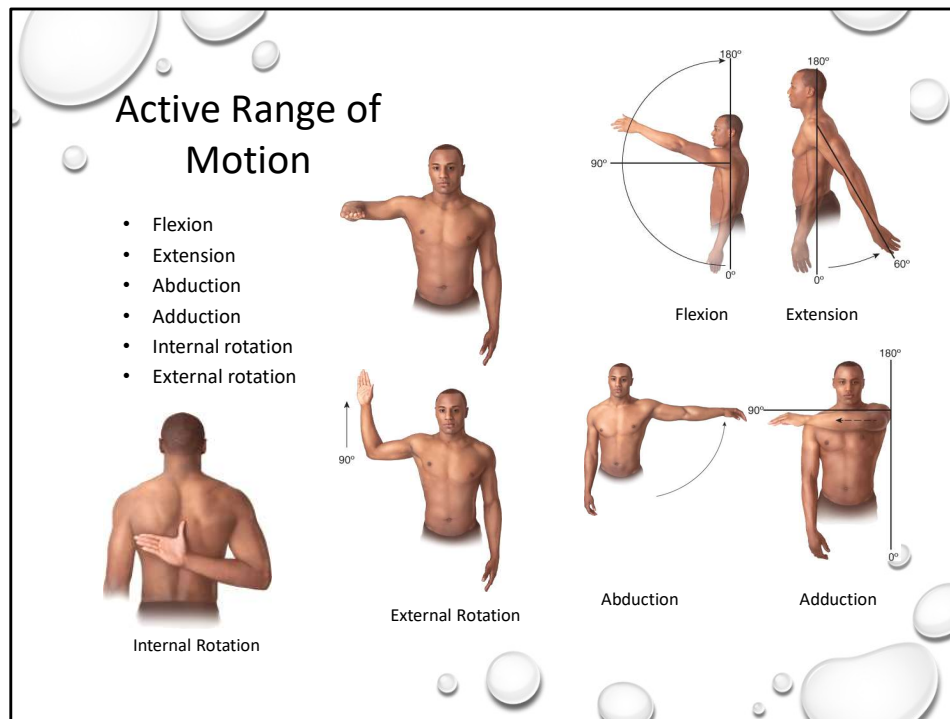


- Sulcus sign



- Apply both superior and inferior traction to AC joint to assess for reduction and/or exaggeration of AC joint separation
- Apply inferior traction to the glenohumeral joint looking for laxity in the joint

This is specific for the shoulder for ac joint separation or laxity at shoulder



Bate's book

Perform passive range of motion when your patient has restricted or limited active range of motion, by patient not activating their muscles you can see if ROM is the same or better.

ABduction

- The full Abduction arc is due to 4 muscles. Pain in a certain part of the arc may help you narrow down where the injury is.
- Supraspinatus – first 15 degrees
- Deltoid- 15- 90 degrees
- Serratus Anterior and Trapezius- 90 degrees to 180 degrees.



Studies now indicate that this is not so cut and dry, that supraspinatus and deltoild overlap with degree of arc.

Apley Scratch Test

- First part: reach overhead to opposite scapula (tests ABduction and External Rotation)
- Second part: reach behind to touch opposite scapula (tests ADDuction and Internal Rotation)

Overall Shoulder Rotation	Apley scratch test. Ask the patient to touch the opposite scapula using the two motions shown below.
	 Tests adduction and internal rotation.
Tests abduction and external rotation.	

Strength
the need
for Neuro
exam

Weakness present?
Isolate the muscle if
possible.



Neuro:

Sensation: C5–T1
dermatomes if cervical
involvement suspected.

Reflexes

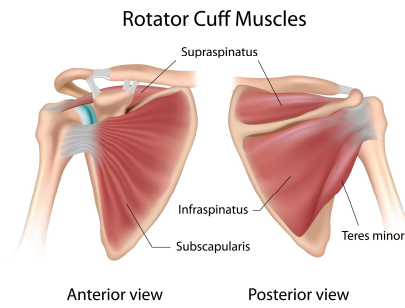
Rotator Cuff

INTRINSIC SPECIAL TESTS

Test the rotator cuff muscles

SITS muscles

- Supraspinatus
- Infraspinatus
- Teres minor
- Subscapularis



Strength Testing

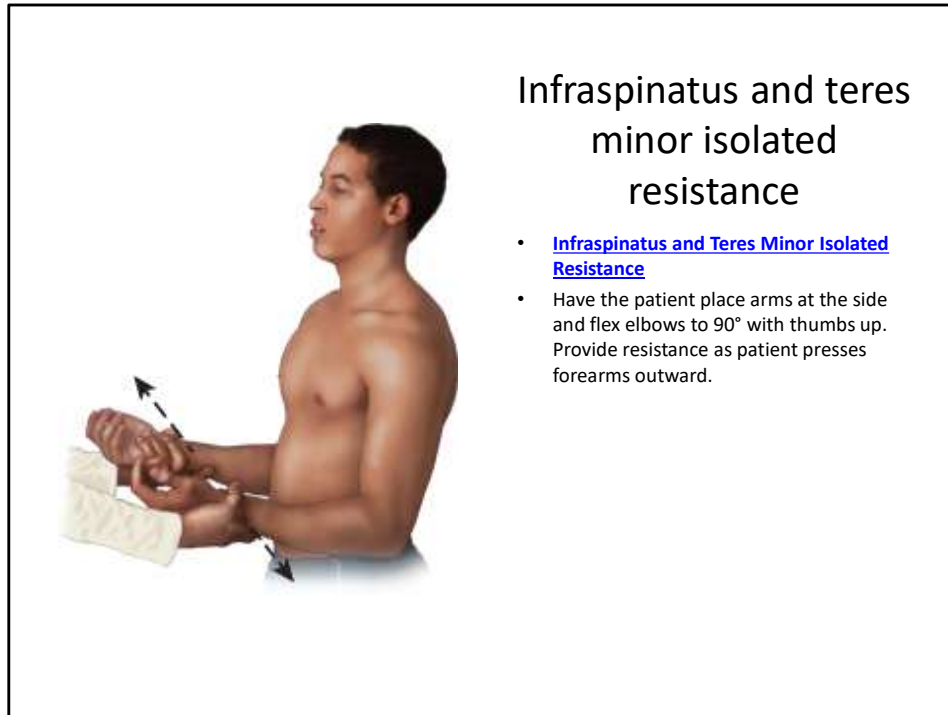
- **Supraspinatus (*Empty Can test or Jobe Test*) -**
Have patient resist downward force with arm held in 90° abduction with 30° forward flexion and internally rotated (thumbs down)
- [Empty Can Test](#)



Drop Arm Test

- Have patient fully abduct arm above their head and lower it slowly. A seamless decent is a normal exam. A sudden drop of the arm indicates supraspinatus tear.
- [Drop Arm Test](#)





If there is weakness present on this exam, it is more commonly an infraspinatus tear and less commonly a teres minor tear.

Strength Testing

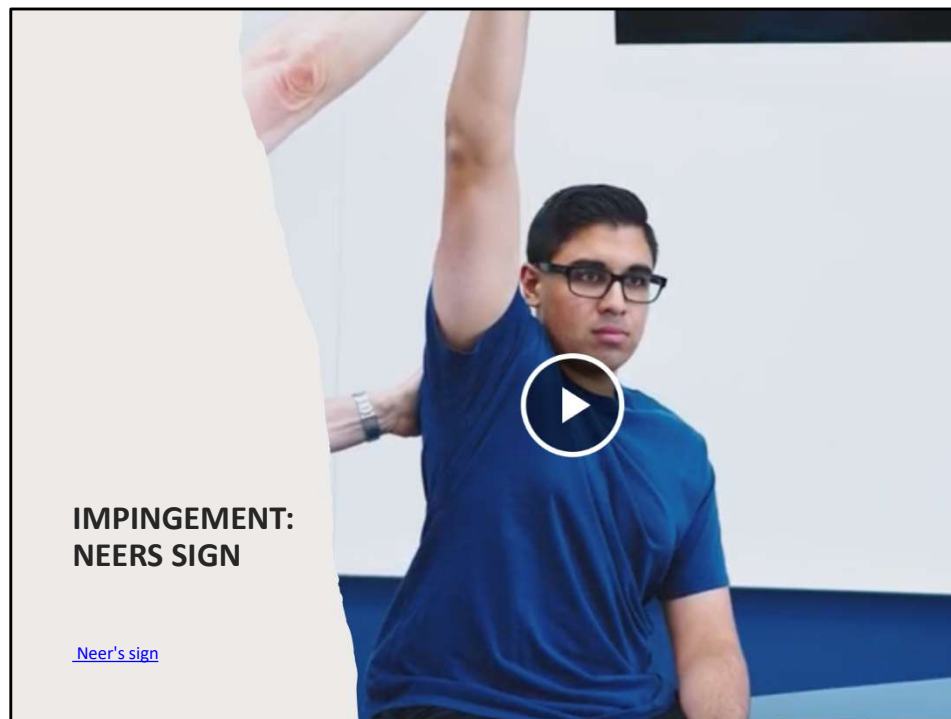
- **Subscapularis**- Have patient place hand behind their back with palm facing away from body. Have patient “lift off” or away from the back against resistance.
- [Lift Off Test](#)



Shoulder Special Tests

Extrinsic tests





Two tests that are usually done in sequence with each other test for impingement underneath the AC joint. Both maneuvers essentially decrease the space under the ac joint, so if tendinitis, bursitis is present it will reproduce the patient's pain.

Hawkins Impingement Sign

- Elevate patient's shoulder to 90°, flex elbow to 90°, place forearm in neutral position. Passively internally rotate arm. Pain with this test suggests rotator cuff tear or impingement syndrome.
- [Hawkin's Test](#)



Crossover Test

- Elevate shoulder to 90° and adduct arm across the body in horizontal plane. Pain over AC joint suggests arthritis of the joint.
- [Cross Arm Test](#)



Shoulder Apprehension Test

- Place patient's arm in 90° abduction, then maximal external rotation. If patient demonstrates apprehension that shoulder will dislocate, this is positive for glenohumeral joint instability.
- [Apprehension Test](#)

Test can be performed in seated or supine position



If apprehension test is positive, then Fowler's relocation test should be performed.

O'Brien's Active Compression Test

- To check for labral tears (SLAP)
- [O'Brien's Active Compression Test](#)

Positive test is when pain is elicited with thumb down and pain relieved with the thumb up.

LOCATION of the pain is important: Pain at top of shoulder- AC joint pathology, Pain in Shoulder- SLAP lesion

Speed's Test

- Pain in the bicipital groove suggests biceps tendonitis or SLAP lesion.
- Weakness - a severe biceps tendon strain may be present.
- [Speed's Test](#)



Yergason's Test

- A positive test: feeling the biceps tendon sublux out of groove (transverse humeral ligament is torn) or if pain, then suspect biceps tendinosis or SLAP lesion.
- [Yergason's Test](#)



Clinical Pearls

Painful active ROM
with normal
passive ROM →
soft tissue

Limited active &
passive ROM →
joint pathology

Night pain may
indicate cuff tear or
malignancy

Summary



- MOST SHOULDER PAIN IS MECHANICAL



- USE HISTORY + IPROMS FOR STRUCTURE-BASED DIAGNOSIS



- ALWAYS CHECK RED FLAGS & REFERRED PAIN SOURCES